1.

Unsalted : 5 mg Na / oz

Lightly Salted : 75 mg Na / oz

Salted : 150 mg Na / oz

$$\frac{5\,mg\ Na}{1\,oz} \cdot \frac{1\,g}{1000\,mg} \cdot \frac{1\,oz}{0.0625\,lbm} = 0.08\ g/lbm$$ ✓

3/4 tsp NaCl

1 lbm peanuts unsalted

1 tsp NaCl = 6g

$$\frac{3}{4}\ tsp\ NaCl \cdot \frac{6g}{1\ tsp\ NaCl} = \frac{9}{2} = 4.5g$$

$$2Na + Cl_2 \rightarrow 2NaCl$$

a)

$$\frac{75\,mg\ Na}{1\,oz} \cdot \frac{1\,g}{1000\,mg} \cdot \frac{1\,oz}{0.0625\,lbm} = 1.2\ g\ Na/lbm$$ ✓

$$\frac{1.2\,g\ Na}{1\,lbm} \cdot \frac{1\,mol\ Na}{22.99\,g\ Na} \cdot \frac{2\,mol\ NaCl}{2\,mol\ Na} \cdot \frac{58.44\,g\ NaCl}{1\,mol\ NaCl} = 3.05\ g\ NaCl/lbm$$

$$\frac{3.05\,g\ NaCl}{1\,lbm} \cdot 1\,lbm = 3.05\ g\ NaCl$$

$$(4.5 + 0.08) - 3.05 = 1.45g\ NaCl$$ ✗

−3 pts

$$1.45g\ NaCl \cdot \frac{1\,tsp}{6g\ NaCl} = 0.242\ tsp\ NaCl\ needed$$ ✗

b)

$$\frac{150\,mg\ Na}{oz} \cdot \frac{1g\ Na}{1000\,mg\ Na} \cdot \frac{1\,oz}{0.0625\,lbm} = 2.4\ g\ Na/lbm$$ ✓

$$\frac{2.4\,g\ Na}{1\,lbm} \cdot \frac{1\,mol\ Na}{22.99\,g\ Na} \cdot \frac{2\,mol\ NaCl}{2\,mol\ Na} \cdot \frac{58.44\,g\ NaCl}{1\,mol\ NaCl} = 6.1\ g\ NaCl/lbm$$

$$\frac{6.1\,g\ NaCl}{1\,lbm} \cdot 1\,lbm = 6.1g\ NaCl$$

No more table salt needed as it has overshot the original amount ✓

no part c, −6 pts

P1: 11/20

2)

$Cu + Cl_2 \rightarrow CuCl_2$

$Cu + \frac{1}{2} Cl_2 \rightarrow CuCl$

$x_{CuCl_2} = \frac{168.047}{200} = 0.84$

$x_{CuCl} = \frac{31.953}{200} = 0.16$

$\frac{x_{CuCl_2}}{x_{CuCl}} = \frac{0.84}{0.16} = 5.25$

5.25 : 1

$m_{Cu_2Cl} + m_{CuCl} = 200$

$x_1 = \frac{m_{Cu}}{m_{CuCl_2}} = \frac{63.55}{134.45} = 0.473$

$x_2 = \frac{m_{Cu}}{m_{CuCl}} = \frac{63.55}{99} = 0.642$

$0.473\, m_{CuCl_2} + 0.642\, m_{CuCl} = 100$

$m_{CuCl_2} = 168.047$

$m_{CuCl} = 31.953$

P2: 20/20

3)

a)

$$\Delta H = \int_{T_1}^{T_2} (A + BT + CT^2 + DT^3)\, dT$$

$$= \left(AT + \tfrac{1}{2}BT^2 + \tfrac{1}{3}CT^3 + \tfrac{1}{4}DT^4\right)\Big|_{300}^{700}$$

$$= (21{,}033.1) - (8{,}568.41)$$

$A = 28.11$
$B = -3.68 \times 10^{-6}$
$C = 1.746 \times 10^{-5}$
$D = -1.065 \times 10^{-8}$

$\Delta H = 12{,}464.7$ J/mol

b) $t = \frac{T}{1000}$ $\quad T = 1000t$

$$C_p(t) = A + B(1000t) + C(1{,}000{,}000t^2) + D(1{,}000{,}000{,}000t^3)$$

$$\Delta H = \int_{300}^{700} C_p(t) = \int_{300}^{700} A + B(1000t) + C(1{,}000{,}000t^2) + D(1{,}000{,}000{,}000t^3)\, dt$$

$$\Delta H = \left(At + \frac{1000}{2}Bt^2 + \frac{1{,}000{,}000}{3}Ct^3 + \frac{1{,}000{,}000{,}000}{4}Dt^4\right)\Big|_{300}^{700}$$

$$\Delta H = (21{,}033.1) - (8{,}568.41)$$

$\Delta H = 12{,}464.7$ J/mol

P3: 20/20

4)

$$\int_1^h \frac{1}{40 - 4\sqrt{x-1}}\, dx = -\frac{\sqrt{h-1}}{2} - 5\log(10 - \sqrt{h-1}) + 5\log(10)$$

$u = \sqrt{x-1}$ $\qquad u(1) = 0$

$u^2 = x-1$ $\qquad u(h) = \sqrt{h-1}$

$2u\,du = dx$

$$= \int_0^{\sqrt{h-1}} \frac{2u}{40-4u}\,du$$

$$= \int_0^{\sqrt{h-1}} \frac{2u}{4(10-u)}\,du$$

$$= \int_0^{\sqrt{h-1}} \frac{u}{2(10-u)}\,du$$

P4: 20/20

$$= \frac{1}{2}\int_0^{\sqrt{h-1}} \frac{u}{10-u}\,du$$

$$= \frac{1}{2}\int_0^{\sqrt{h-1}} -1 + \frac{10}{10-u}\,du$$

$w = 10-u$

$dw = -du$

$$= \frac{1}{2}\left(\int_0^{\sqrt{h-1}} -1\,du + \int_0^{\sqrt{h-1}} \frac{10}{10-u}\,du\right)$$

$$= \frac{1}{2}\left((-u)\Big|_0^{\sqrt{h-1}} + \int_0^{\sqrt{h-1}} \frac{10}{w}\,du\right)$$

$$= \frac{1}{2}\left(-\sqrt{h-1} + (10\ln(10-u))\Big|_0^{\sqrt{h-1}}\right)$$

$$= \frac{1}{2}\left(-\sqrt{h-1} + 10\ln(10-\sqrt{h-1}) - 10\ln(10)\right)$$

$$= -\frac{1}{2}\sqrt{h-1} + 5\ln(10-\sqrt{h-1}) - 5\ln(10)$$

5)

C_5H_6OH

7 Kcal/g

$\Delta H_g = -1{,}235$ KJ/mol

$\Delta H_\ell = -1{,}367$ KJ/mol

$T = 25°c$

$P = 1atm$

molar mass of ethanol = 46.07 g/mol

Kcal/g to KJ/mol

$$\Delta H = \frac{7\text{ Kcal}}{1g} \cdot \frac{4.184\text{ KJ}}{1\text{ Kcal}} \cdot \frac{46.07\text{ g}}{1\text{ mol}} = 1{,}349.3\ \frac{KJ}{mol}$$

I can conclude this mixture of ethonal is likely to be a liquid State since the magnitude of ΔH_ℓ and ΔH are closer in value than ΔH_g

P5: 20/20